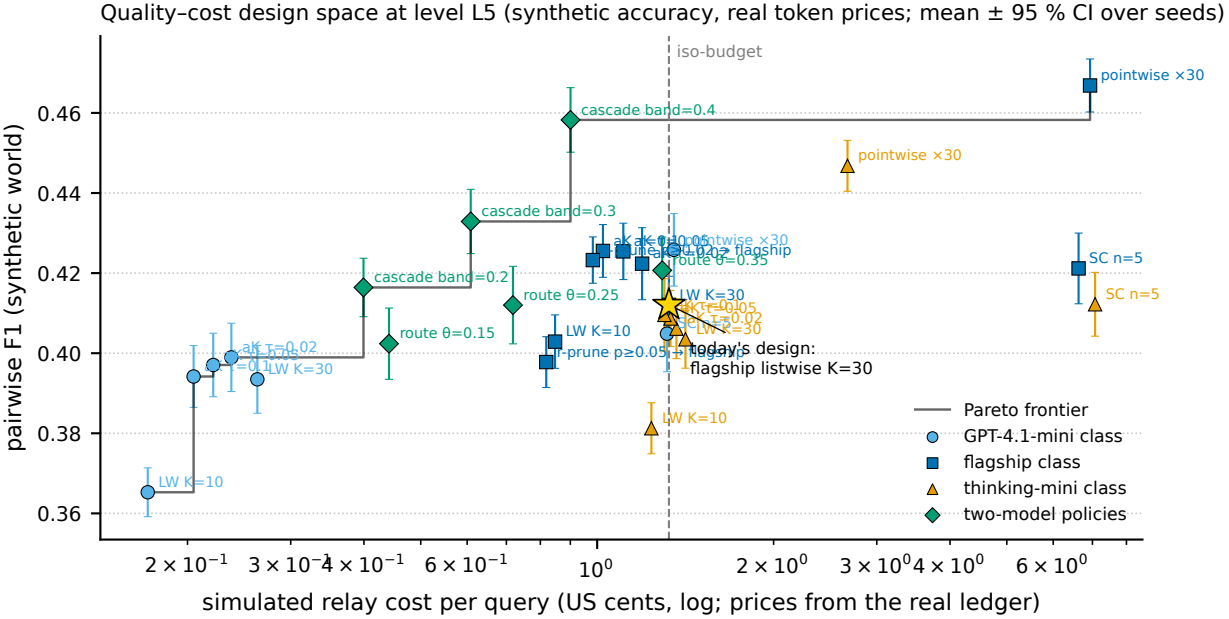
Quality–cost design space at level L5 (synthetic accuracy, real token prices; mean ± 95 % CI over seeds)
iso-budget
pointwise ×30
cascade band=0.4
pointwise ×30
cascade band=0.3
SC n=5
cascade band=0.2
SC n=5
route θ=0.25
route θ=0.15
LW K=10
LW K=30
LW K=10
today's design:
flagship listwise K=30
LW K=10
Pareto frontier
GPT-4.1-mini class
flagship class
thinking-mini class
two-model policies
pairwise F1 (synthetic world)
0.46
0.44
0.42
0.40
0.38
0.36
2 × 10⁻¹
3 × 10⁻¹
4 × 10⁻¹
6 × 10⁻¹
10⁰
2 × 10⁰
3 × 10⁰
4 × 10⁰
6 × 10⁰
simulated relay cost per query (US cents, log; prices from the real ledger)